"Same Concepts Better You!

Practice
Analyze
Improve
Repeat

- [x] Learn Concepts
- [x] Solve Questions
- [x] Boost Accuracy
- [x] Crack GATE!

GATE Aptitude Cheat Sheet

Complete Guide

Aptitude Builds Your Edge!



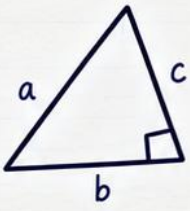
Prep Smart Score High!

Small Steps Big Results

12 Pages • All Key Concepts • Quick Revision

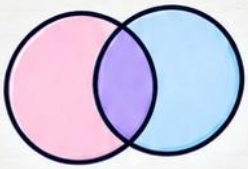


Numbers Make Sense!



Formulas Save Time!

1	Numerical Ability
2	Engineering Mathematics
3	Verbal Ability
4	Data Interpretation
5	Logical Reasoning
6	Analytical Reasoning
7	Set Theory
8	Geometry & Mensuration
9	Basic Statistics
10	Formulas & Shortcuts
11	Previous-Year Patterns
12	Top Practice Questions



Logic Connects Everything!



More Practice Less Stress!





Same Goal Higher Score!

"Aptitude Today Opportunities Tomorrow"

GATE Prep Better Tomorrow!

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1. Quantitative Aptitude Basics + Number System

Small Steps Big Progress!

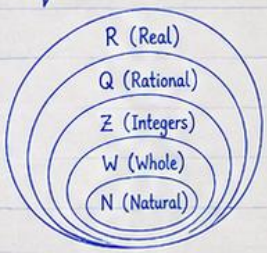
1. What is Quantitative Aptitude ?

- Quantitative Aptitude (QA) tests your ability to solve numerical problems using logic, concepts and calculations.
- It includes topics like Number System, Arithmetic, Algebra, Geometry, Data Interpretation etc.
- Common in competitive exams like GATE, CAT, GRE, placements etc.
- Key skills : concept clarity + practice + speed.

"Practice turns confusion into confidence."

2. Types of Numbers

- Natural Numbers (N) : 1, 2, 3, 4, ... (counting numbers)
- Whole Numbers (W) : 0, 1, 2, 3, ... (includes 0)
- Integers ($\mathbb{Z}$) : ..., -2, -1, 0, 1, 2, ...
- Rational Numbers (Q) : Numbers of the form $\frac{p}{q}$ $(q \neq 0)$
  e.g. $\frac{1}{2}, -\frac{3}{4}$, 5, 0.125
- Irrational Numbers : Cannot be written as $\frac{p}{q}$
  e.g. $\sqrt{2}, \sqrt{3}, \pi$, 0.101001...
- Real Numbers (R) : All rational and irrational numbers.



3. Even and Odd Numbers

- Even : Divisible by 2. (last digit 0, 2, 4, 6, 8)
  e.g. 14, 26, 100
- Odd : Not divisible by 2. (last digit 1, 3, 5, 7, 9)
  e.g. 15, 27, 101

Even number = $2n$, Odd number = $2n + 1$
(where n is an integer)

4. Prime and Composite Numbers

- Prime : A number > 1 with exactly two factors (1 and itself).
  e.g. 2, 3, 5, 7, 11, ...
- Composite : A number > 1 with more than two factors.
  e.g. 4 (1,2,4), 6 (1,2,3,6)

Note: 1 is neither prime nor composite.

5. Factors and Multiples

- Factors : Numbers that divide a number completely.
  e.g. Factors of 12 → 1, 2, 3, 4, 6, 12
- Multiples : Numbers obtained by multiplying a number with natural numbers.
  e.g. Multiples of 5 → 5, 10, 15, 20, ...

6. Divisibility Rules (Common)

2	Last digit is 0, 2, 4, 6 or 8
3	Sum of digits is divisible by 3
4	Last two digits are divisible by 4
5	Last digit is 0 or 5
6	Divisible by both 2 and 3
8	Last three digits are divisible by 8
9	Sum of digits is divisible by 9
10	Last digit is 0
11	(Sum of digits at odd places - sum at even places) is divisible by 11

7. HCF and LCM

- HCF (Highest Common Factor)
  Largest number that divides two or more numbers.
  e.g. HCF(12, 18) = 6
- LCM (Least Common Multiple)
  Smallest number that is a multiple of given numbers.
  e.g. LCM(12, 18) = 36
- Relation : For two numbers a, b

$HCF(a, b) \times LCM(a, b) = a \times b$

8. Remainder Basics

- If a number N is divided by d,
  $N = d \times q + r$
  q → quotient
  r → remainder $(0 \leq r < d)$
- Example : $29 \div 5$
  $29 = 5 \times 5 + 4$
  Remainder = 4

Useful in problems related to units digit, divisibility and cyclic patterns.

9. Fractions, Decimals and Percentage

- Fraction : Part of a whole. e.g. $\frac{3}{4}$
- Decimal : Another form of fraction. e.g. $0.75 = \frac{3}{4}$
- Percentage : Means per hundred. e.g. $25\% = \frac{25}{100} = \frac{1}{4}$
- Conversion : $1\% = \frac{1}{100} = 0.01$
  $x\%$ of a number $= \left(\frac{x}{100}\right) \times$ number

10. Squares and Cubes (Quick Facts)

Squares (n^2)	Cubes (n^3)
$1^2 = 1$	$1^3 = 1$
$2^2 = 4$	$2^3 = 8$
$3^2 = 9$	$3^3 = 27$
$4^2 = 16$	$4^3 = 64$
$5^2 = 25$	$5^3 = 125$
$6^2 = 36$	$6^3 = 216$
$7^2 = 49$	$7^3 = 343$
$8^2 = 64$	$8^3 = 512$
$9^2 = 81$	$9^3 = 729$
$10^2 = 100$	$10^3 = 1000$

11. GATE Tip / Shortcuts

- ✔ Learn and remember squares, cubes and common divisibility rules.
- ✔ Use approximation to save time in calculations.
- ✔ Check options (if any) before full calculation.
- ✔ Practice previous year questions.
- ✔ Improve speed with regular mock tests.

"Concepts + Practice + Consistency = Results"

Keep Learning!

Same Concepts Bigger Results !

2. Percentage + Ratio + Proportion + Averages

Practice Solve Improve Repeat !

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1. Percentage

- Percentage means "per hundred".
- $x\% = \frac{x}{100}$
- Used to compare quantities easily.

Percentage helps in quick comparison.

Common Conversions

- $1\% = \frac{1}{100} = 0.01$
- $5\% = 0.05$
- $10\% = 0.1$
- $25\% = 0.25$
- $50\% = 0.5$
- $75\% = 0.75$
- $100\% = 1$

2. Percentage Formulas

- Find $x\%$ of a number $= \frac{x}{100} \times N$
- Find what percent A is of B $= \frac{A}{B} \times 100$
- Increase by $x\% = N\left(1 + \frac{x}{100}\right)$
- Decrease by $x\% = N\left(1 - \frac{x}{100}\right)$
- Successive change $= \left(1 \pm \frac{x}{100}\right)\left(1 \pm \frac{y}{100}\right) - 1$

Example :

Find 20% of 250

$= \frac{20}{100} \times 250$

$= 50$

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3. Ratio

- Ratio compares two quantities.
- If ratio of a : b, it means $\frac{a}{b}$ or a : b.
- Can be simplified by dividing both terms by their GCD.

Example : Simplify 18 : 24

GCD = 6

18 : 24 = 3 : 4

Important Points

- $a : b = ka : kb$
- $a : b = \frac{a}{b}$
- If $a : b = m : n$, then $a = km$, $b = kn$.
- Ratio has no unit.
- Order matters $(a : b \neq b : a)$

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4. Proportion

- Proportion means two ratios are equal.
- If $\frac{a}{b} = \frac{c}{d}$, then $a : b = c : d$.
- Product of extremes = Product of means.

If $a : b = c : d \rightarrow ad = bc$

Example : If $2 : 3 = x : 9$

$2 \times 9 = 3 \times x$

$18 = 3x \rightarrow x = 6$

Properties

- If $a : b = b : c$, then $a : b : c$ is in continued proportion.
- If $a : b = c : d$, then $(a + b) : b = (c + d) : d$

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5. Averages (Mean)

- Average $= \frac{\text{Sum of all values}}{\text{Number of values}}$
- Used to find the central value of a data set.
- If n numbers have average A, then sum = nA.
- If a new number x is added, new average $= \frac{nA + x}{n + 1}$

Example : Average of 10, 20, 30, 40

$= \frac{10 + 20 + 30 + 40}{4} = 25$

6. Weighted Average

- Used when different values have different weights.
- Weighted Average $= \frac{\sum (w_i \times x_i)}{\sum w_i}$

Example : Marks: 80 (weight 2), 70 (weight 1)

Weighted Average $= \frac{80 \times 2 + 70 \times 1}{2 + 1}$

$= \frac{160 + 70}{3} = 76.67$

7. Combined Average

- If group 1 has n_1 items with average A_1 and group 2 has n_2 items with average A_2,
- Combined Average $= \frac{n_1 A_1 + n_2 A_2}{n_1 + n_2}$

Example : Class A: 30 students, avg = 70

Class B: 20 students, avg = 80

Combined Average $= \frac{30 \times 70 + 20 \times 80}{30 + 20}$

$= \frac{2100 + 1600}{50} = 74$

8. Quick Tips & Common Mistakes

Tips

- Convert percentages to fractions for easy calculation.
- Simplify ratios to lowest terms.
- Use cross multiplication in proportions.
- Check if increase or decrease is on the original value or new value.
- In averages, keep track of total sum and count.
- Use approximation for faster calculation in exams like GATE.

Common Mistakes

- Confusing % increase vs % decrease.
- Not simplifying ratios.
- Wrong order in proportion.
- Forgetting weights in weighted average.
- Mixing units.
- Rounding errors in quick calculations.
- Not checking options (if given).

Small Concepts Big Progress !

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Keep Practicing !

Learn Practice Improve Get Ahead !

3. Profit & Loss + Simple & Compound Interest

Same Concepts Bigger Opportunities !

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1. Profit & Loss Basics

- Profit = SP − CP
- Loss = CP − SP
- Profit % = $\left(\frac{\text{Profit}}{\text{CP}}\right) \times 100$
- Loss % = $\left(\frac{\text{Loss}}{\text{CP}}\right) \times 100$
- SP = CP $\left(1 + \frac{\text{Profit \%}}{100}\right)$
- SP = CP $\left(1 - \frac{\text{Loss \%}}{100}\right)$
- CP = $\frac{\text{SP}}{\left(1 + \frac{\text{Profit \%}}{100}\right)}$ or $\frac{\text{SP}}{\left(1 - \frac{\text{Loss \%}}{100}\right)}$

Example :
A book is bought for ₹200 and sold for ₹250. Find profit %.

Profit = 250 − 200 = 50

Profit % = $\frac{50}{200} \times 100$

= 25%

2. Successive Profit & Loss / Discount

- If two successive gains of a% and b% are given, net gain % = $\left(a + b + \frac{ab}{100}\right)\%$
- If two successive losses of a% and b% are given, net loss % = $\left(a + b - \frac{ab}{100}\right)\%$
- If a gain of a% and a loss of a% are given, net change % = $\left(-\frac{a^2}{100}\right)\%$ (always a loss)
- Successive discounts of a% and b% = $\left(a + b - \frac{ab}{100}\right)\%$

Example :
A price is increased by 20% and then decreased by 10%. Find net change %.

Net change = $20 - 10 - \frac{20 \times 10}{100}$

= 30 − 2

= 18%

(Overall gain)

3. Marked Price, Discount and Selling Price

- Marked Price (MP) : The printed price.
- Discount = MP − SP
- Discount % = $\frac{\text{Discount}}{\text{MP}} \times 100$
- SP = MP $\left(1 - \frac{\text{Discount \%}}{100}\right)$
- If successive discounts, use the formula: $\left(a + b - \frac{ab}{100}\right)\%$

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Example :
A shirt is marked at ₹800 and is sold at 25% discount. Find the selling price.

SP = $800\left(1 - \frac{25}{100}\right)$

= $800 \times \frac{75}{100}$

= ₹600

4. Simple Interest (SI)

- SI = $\frac{P \times R \times T}{100}$
- P = Principal amount
- R = Rate of interest (per annum)
- T = Time (in years)
- Amount (A) = P + SI
- SI is directly proportional to P, R and T.

Example :
Find the simple interest on ₹5000 at 8% per annum for 3 years.

SI = $\frac{5000 \times 8 \times 3}{100}$

= 1200

Amount = 5000 + 1200

= ₹6200

5. Compound Interest (CI)

- A = P $\left(1 + \frac{R}{100}\right)^T$
- CI = A − P
- P = Principal amount
- R = Rate of interest (per annum)
- T = Time (in years)
- A = Amount after T years

Example :
Find the compound interest on ₹2000 at 10% per annum for 2 years.

A = $2000\left(1 + \frac{10}{100}\right)^2$

= $2000\,(1.1)^2$

= 2000 × 1.21 = ₹2420

CI = 2420 − 2000 = ₹420

6. CI for Different Compounding Periods

Compounding Period	Amount (A)
Annually	P $\left(1 + \frac{R}{100}\right)^T$
Half-yearly	P $\left(1 + \frac{R}{200}\right)^{2T}$
Quarterly	P $\left(1 + \frac{R}{400}\right)^{4T}$
Monthly	P $\left(1 + \frac{R}{1200}\right)^{12T}$

Note :
- As the compounding frequency increases, CI increases.
- For small time periods (T < 1 year), convert time into the same unit as the compounding period.

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7. Important Shortcuts & Concepts

1	If SP = k × CP, then Profit % = (k − 1) × 100
2	If SP = $\frac{CP}{k}$, then Loss % = $\left(1 - \frac{1}{k}\right) \times 100$
3	If profit % and loss % are equal, net result is always a loss.
4	For SI, if rate is doubled, SI doubles.
5	For CI, amount grows exponentially (not linearly).
6	Use approximation for large powers in exams.
7	In case of equal installments, total interest can be calculated using average principal (if needed).

8. Common Mistakes to Avoid

- ✗ Confusing CP and SP.
- ✗ Using simple interest formula for compound interest.
- ✗ Forgetting to convert percentage to decimal.
- ✗ Not reading the time period carefully (months vs years).
- ✗ Ignoring compounding frequency (half-yearly, quarterly).
- ✗ Calculation errors in successive profit/loss or discounts.
- ✗ Assuming profit % and loss % cancel each other (they don't).

Consistent Practice Brings Results !

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Small Steps Big Success !

Learn Solve Practice Get Better !

4. Time, Speed & Distance + Work & Pipes

Same Concepts Bigger Opportunities !

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1. Time, Speed & Distance Basics

- Speed (S) = $\frac{\text{Distance (D)}}{\text{Time (T)}}$
- Distance (D) = S x T
- Time (T) = $\frac{D}{S}$
- Units : Keep units consistent (km-h, m-s etc.)
- Average Speed = $\frac{\text{Total Distance}}{\text{Total Time}}$

Example :
A car travels 180 km in 3 hours. Find speed.

$S = \frac{D}{T} = \frac{180}{3}$

$= 60$ km/h

2. Relative Speed

- In same direction :
  Relative speed = $|S_1 - S_2|$
- In opposite direction :
  Relative speed = $S_1 + S_2$

Example :
Two trains of 60 km/h and 80 km/h move in opposite directions.

Relative speed = 60 + 80

= 140 km/h

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3. Problems Based on Time

- If a person covers a distance in T hours at speed S, time taken at new speed S' :

$T' = T \times \frac{S}{S'}$

- If distance is same and speed increases by x%,
  new time = $T \times \frac{100}{100 + x}$

Example :
A person takes 4 hours to complete a journey. If speed is increased by 25%, find new time.

$T' = \frac{4 \times 100}{125}$

$= 3.2$ hours

4. Average Speed

- If equal distances are covered at speeds S_1 and S_2 :
  Average speed = $\frac{2 S_1 S_2}{S_1 + S_2}$
- If equal times are spent at speeds S_1 and S_2 :
  Average speed = $\frac{S_1 + S_2}{2}$

Example :
A car covers first half distance at 40 km/h and second half at 60 km/h.

Average speed = $\frac{2 \times 40 \times 60}{40 + 60}$

$= \frac{4800}{100}$

= 48 km/h

5. Work and Time Basics

- Work = Efficiency x Time
- If A can do a work in x days,
  Efficiency of A = $\frac{1}{x}$
- If A and B work together,
  Efficiency = $\frac{1}{x} + \frac{1}{y}$
- Time together = $\frac{1}{\frac{1}{x} + \frac{1}{y}} = \frac{xy}{x + y}$

Example :
A can do a work in 6 days and B in 8 days. In how many days will they complete it together?

$T = \frac{6 \times 8}{6 + 8}$

$= \frac{48}{14}$

$= \frac{24}{7}$ days

6. Work Related Problems

- If A is x% more efficient than B,
  Efficiency of A = $\frac{100 + x}{100}$ x (Efficiency of B)
- If A can do a work in x days and B in y days,
  A alone in t days = (work done by A in t days)

$= \frac{t}{x}$

Remaining work = $1 - \left(\frac{t}{x}\right)$

Example :
A can do a work in 10 days and B is 20% more efficient than A. In how many days can B complete the work?

Efficiency of A = $\frac{1}{10}$

Efficiency of B = $\frac{1}{10} \times 1.2$

$= \frac{1}{8}$

So, B can do the work in 8 days.

7. Pipes and Cisterns Basics

- Inlet pipe : Fills the tank (positive work)
- Outlet pipe : Empties the tank (negative work)
- Net work = (Sum of inlet efficiencies) – (Sum of outlet efficiencies)
- Time to fill/empty = $\frac{1}{\text{Net work}}$



8. Pipes Example

Example :
A pipe can fill a tank in 12 hours and another pipe can empty it in 18 hours. If both are opened together, in how many hours will the tank be full?

Net work = $\frac{1}{12} - \frac{1}{18}$

$= \frac{3 - 2}{36} = \frac{1}{36}$

Time = $\frac{1}{\frac{1}{36}} = 36$ hours

9. Quick Tips

- Keep units consistent (km/h, m/s).
- Draw a diagram for relative motion and pipes problems.
- For work problems, use efficiency (1 / time) method.
- Convert percentages to fractions quickly.
- Check for special cases (equal distance, equal time).
- Practice previous year questions to identify patterns.

Practice Today Crack Tomorrow !

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Small Steps Big Results !

Learn
Practice
Improve
Get Ahead !

5. Algebra + Equations + Progressions

Same
Concepts
Bigger
Opportunities !

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1. Algebra Basics

- Algebra uses variables (like x, y) to represent unknown values.
- Follow the rules of operations : BODMAS (Bracket, Order, Division, Multiplication, Addition, Subtraction).
- Simplify expressions before solving.

Example : Simplify $2(x+3) - 4(x-1)$

$= 2x + 6 - 4x + 4$

$= -2x + 10$

2. Common Algebraic Identities

1. $(a+b)^2 = a^2 + 2ab + b^2$
2. $(a-b)^2 = a^2 - 2ab + b^2$
3. $a^2 - b^2 = (a+b)(a-b)$
4. $(x+a)(x+b) = x^2 + (a+b)x + ab$
5. $(x-a)(x-b) = x^2 - (a+b)x + ab$
6. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
7. $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
8. $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

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3. Equations (Linear, Quadratic)

Linear Equation

- General form : $ax + b = 0$
- Solution : $x = -\frac{b}{a}$
- Example : $3x + 6 = 0 \Rightarrow x = -2$

Quadratic Equation

- General form : $ax^2 + bx + c = 0$
- Roots : $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Discriminant $(D) = b^2 - 4ac$
  - $D > 0 \rightarrow$ two real roots
  - $D = 0 \rightarrow$ equal roots
  - $D < 0 \rightarrow$ no real roots

Example : $x^2 - 5x + 6 = 0$

$(x-2)(x-3) = 0 \Rightarrow x = 2, 3$

4. Simultaneous Equations

- Two or more equations with same variables.
- Solve using substitution, elimination or matrix method.

Example :

$2x + 3y = 12$

$x - y = 1$

From second, $x = y + 1$

Substitute in first :

$2(y+1) + 3y = 12$

$5y = 10 \Rightarrow y = 2$

$x = 3$

5. Inequalities

- Represents a range of values.
- Common signs : $>$, $<$, $\geq$, $\leq$
- Solve like equations, but be careful when multiplying or dividing by a negative number (inequality sign reverses).

Example : Solve $2x - 5 > 7$

$2x > 12 \Rightarrow x > 6$

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6. Modulus (Absolute Value)

- $|x| = x$ (if $x \geq 0$)
- $|x| = -x$ (if $x < 0$)
- Used in equations and inequalities.

Example : Solve $|x - 3| = 5$

$x - 3 = 5$ or $x - 3 = -5$

$x = 8$ or $x = -2$

7. Arithmetic Progression (AP)

- Numbers with a constant difference (d).
- General term : $a_n = a + (n-1)d$
- Sum of first n terms : $S_n = \frac{n}{2}[2a + (n-1)d]$

Example : First term = 3, d = 2, n = 10

$S_{10} = \frac{10}{2}[2(3) + 9(2)]$

$= 5[6 + 18] = 5 \times 24 = 120$

8. Geometric Progression (GP)

- Numbers with a constant ratio (r).
- General term : $a_n = ar^{n-1}$
- Sum of first n terms $(r \neq 1)$:

  $S_n = \frac{a(r^n - 1)}{r - 1}$
- Sum to infinity $(|r| < 1)$:

  $S_\infty = \frac{a}{1 - r}$

Example : $a = 2$, $r = 3$, $n = 4$

$S_4 = \frac{2(3^4 - 1)}{3 - 1} = \frac{2(81 - 1)}{2} = 80$

9. Special Progression & Series

- Sum of first n natural numbers :

  $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$
- Sum of squares : $1^2 + 2^2 + \ldots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- Sum of cubes : $1^3 + 2^3 + \ldots + n^3 = \left[\frac{n(n+1)}{2}\right]^2$

Example : Find sum of first 10 natural numbers

$S = \frac{10(11)}{2} = 55$

10. Common Tricks & Tips

- ✓ Simplify expressions before solving.
- ✓ Use factorization wherever possible.
- ✓ Check for common patterns in equations.
- ✓ For large numbers, use AP/GP formulas instead of adding manually.
- ✓ In inequalities, remember to reverse the sign when multiplying/dividing by a negative number.
- ✓ Practice previous year questions to identify patterns.

11. Common Mistakes to Avoid

- ✗ Forgetting BODMAS rules.
- ✗ Sign errors while expanding or factorizing.
- ✗ Not checking both cases in modulus problems.
- ✗ Wrongly applying formulas for AP/GP.
- ✗ Ignoring the condition $r \neq 1$ in GP sum formula.
- ✗ Not verifying the solution in simultaneous equations.
- ✗ Calculation mistakes in long expressions.

Practice
Build Concepts
See Results !

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Small
Steps
Big Success !

More Practice Brighter Results !

6. Permutation, Combination + Probability

Think Solve Apply Succeed !

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1. Fundamental Counting Principle

- If an event can occur in m ways and another event can occur in n ways, then both can occur in $m \times n$ ways.

Example : A person has 3 shirts and 4 pants. In how many ways can he dress?

Total ways = $3 \times 4 = 12$

2. Permutation (Order Matters)

- Arrangement of objects in a particular order.
- $^nP_r = \dfrac{n!}{(n-r)!}$

Special Cases :
- $^nP_n = n!$
- $^nP_1 = n$
- $0! = 1$

Example : In how many ways can 5 students be arranged in a row?

$^5P_5 = 5! = 120$

3. Permutation with Repetition

- If objects are not distinct (some repeated). $\dfrac{n!}{(n_1!\,n_2!\ldots n_k!!}$

Example : In how many ways can the word BALLOON be arranged? (8! letters, with L = 2, O = 2)

$= \dfrac{8!}{2!\,2!} = \dfrac{40320}{4} = 10080$

4. Combination (Order Doesn't Matter)

- Selection of objects without considering order.
- $^nC_r = \dfrac{n!}{r!\,(n-r)!}$

Properties :
- $^nC_r = {}^nC_{n-r}$
- $^nC_0 = {}^nC_n = 1$
- $^nC_1 = n$
- $^nC_2 = \dfrac{n(n-1)}{2}$

Example : In how many ways can 3 students be selected from 7?

$^7C_3 = \dfrac{7!}{3!\,4!} = 35$

5. Combination with Repetition

- If an item can be chosen multiple times.

$$^{n+r-1}C_r = \frac{(n+r-1)!}{r!\,(n-1)!}$$

Example : In how many ways can 4 ice creams be chosen from 3 flavors (repetition allowed)?

$^{3+4-1}C_4 = {}^6C_4 = 15$

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6. Important Identities

1. $^nP_r = {}^nC_r \times r!$
2. $^nC_r = {}^nC_{n-r}$
3. $^nP_r = \dfrac{n!}{(n-r)!}$
4. $^nC_0 = {}^nC_n = 1$
5. $^nC_1 = n, \quad ^nC_2 = \dfrac{n(n-1)}{2}$
6. $^nP_0 = 1$
7. $(n+1)C_n = (n+1)C_{n-1} + nC_n$ (Pascal's Identity)

7. Basics of Probability

- Probability (P) = $\dfrac{\text{Favourable outcomes}}{\text{Total outcomes}}$
- $0 \le P(E) \le 1$
- $P(E) + P(\bar{E}) = 1$ (Complementary event)
- If all outcomes are equally likely, $P(E) = \dfrac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

8. Important Rules

- Addition Rule : $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- For mutually exclusive events $(A \cap B = 0)$: $P(A \cup B) = P(A) + P(B)$
- Multiplication Rule : $P(A \cap B) = P(A) \times P(B|A)$
- For independent events : $P(A \cap B) = P(A) \times P(B)$

9. Conditional Probability

- $P(A|B) = \dfrac{P(A \cap B)}{P(B)} \quad [P(B) \neq 0]$
- $P(B|A) = \dfrac{P(A \cap B)}{P(A)}$
- Independent events : $P(A|B) = P(A)$
- Dependent events : $P(A|B) \neq P(A)$

10. Bayes' Theorem (Useful in GATE)

- If $\{A_1, A_2, \ldots, A_n\}$ are mutually exclusive and exhaustive events, then

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum (P(B|A_j)P(A_j)}$$

Used in problems related to conditional probability, testing, and real-life scenarios.

11. Common Probability Distributions

- Binomial Distribution : $P(X = r) = {}^nC_r\, p^r\, q^{n-r} \quad (q = 1-p)$

  Mean = np, Variance = npq
- If only two outcomes (success/failure) and trials are independent.

12. Tips & Tricks

- ✅ Identify whether order matters.
- ✅ Use nC_r for selection, nP_r for arrangement.
- ✅ Check for mutually exclusive or independent events.
- ✅ Draw a tree diagram for complex problems.
- ✅ Use complementary events $(P(\bar{E}) = 1 - P(E))$.
- ✅ In GATE, probability questions often come from real-life scenarios.

Practice Solves Everything !

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Small Steps Big Success !

Analyze Practice Improve Score Higher !

7. Data Interpretation + Data Sufficiency

Same Concepts Bigger Opportunities !

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1. Data Interpretation (DI) - Overview

- DI is the process of analyzing numerical data (tables, bar graphs, pie charts, line graphs etc.) to extract useful information.
- It tests calculation skills, logical reasoning and interpretation ability.
- Focus on accuracy + speed + approximation.
- Always read the given information carefully.

Tip : Understand the data, don't just look at the numbers. Identify what is asked !

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2. Types of DI

- Tabular DI
- Bar Graph DI
- Line Graph DI
- Pie Chart DI
- Mixed Graph DI
- Caselet DI (Combination)

Steps to Solve

1. Read the question carefully
2. Understand the data/graph
3. Identify what is required
4. Do quick calculations
5. Re-check the answer

3. Important Formulas for DI

- Percentage = $\frac{\text{Part}}{\text{Whole}} \times 100$
- Growth % = $\frac{(\text{New} - \text{Old})}{\text{Old}} \times 100$
- Average = $\frac{\text{Sum of values}}{\text{Number of values}}$
- Ratio = $a : b = \frac{a}{b}$
- Total = Sum of all parts
- If percentage is given, value = $\frac{\% \times \text{Total}}{100}$

Tip : Use approximation to save time in the exam.

4. Bar Graph Interpretation

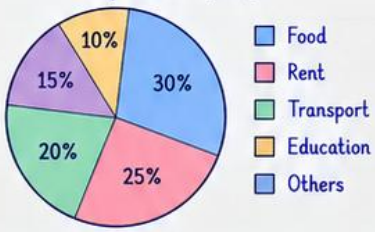


Example Question :

What is the ratio of students in CSE to ME?

80 : 40 = 2 : 1

5. Pie Chart Interpretation



Example Question :

If total expenditure is ₹20,000, how much is spent on Education?

Education = 15%

$= \frac{15}{100} \times 20000$

= ₹3000

6. Line Graph Interpretation



Example Question :

In which year was the maximum sales?

Ans : 2023 (₹80 Lakhs)

What is the increase in sales from 2021 to 2023?

80 - 35 = 45 Lakhs

7. Tabular DI Interpretation

Year	A	B	C	D
2019	45	30	25	20
2020	50	35	30	25
2021	60	40	35	30
2022	65	50	45	40
2023	70	55	50	45

Example Question :

What is the total of A in all years?

45 + 50 + 60 + 65 + 70 = 290

8. Data Sufficiency (DS) - Overview

- Data Sufficiency questions check whether the given statements are sufficient to answer a question.
- You don't need to find the exact answer, just check if the data is sufficient.
- Common in GATE like exams (1 or 2 marks).

Key Idea : Use only the given statements, not your own assumptions.

9. How to Answer Data Sufficiency

1. Read the question carefully.
2. Analyze Statement 1 (S1) alone.
3. Analyze Statement 2 (S2) alone.
4. Analyze both statements together.
5. Choose the correct option.

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10. Typical Answer Options

- **A** S1 alone is sufficient, but S2 alone is not sufficient.
- **B** S2 alone is sufficient, but S1 alone is not sufficient.
- **C** Both S1 and S2 together are sufficient but neither S1 nor S2 alone is sufficient.
- **D** Each statement alone is sufficient.
- **E** Even both statements together are not sufficient.

11. Example (Data Sufficiency)

Q. What is the value of x?

S1 : x + 5 = 10

S2 : x × 2 = 20

Solution :

S1 alone → sufficient (x = 5)

S2 alone → sufficient (x = 10)

Answer : D (Each statement alone is sufficient)

12. Tips & Common Mistakes

- ✓ Focus on whether the data is sufficient, not the answer.
- ✓ Avoid using extra assumptions.
- ✓ Check all possibilities.
- ✓ Practice previous year questions.
- ✓ Manage time, don't over-calculate.
- ✓ Be careful with units, conditions and hidden information.

Better Preparation Brighter Future !

Keep Learning Keep Growing !

Think
Analyze
Solve
Level Up !

8. Logical Reasoning + Analytical Reasoning

Practice
Patterns
Improve
Crack GATE !

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1. Logical Reasoning - Overview

- Tests your ability to think logically, identify patterns and draw valid conclusions.
- Common types: Statements & Conclusions, Assumptions, Arguments, Cause-Effect, Strengthen/Weaken, Analogies, etc.
- No fixed formula, but follows logical rules.

Key Tip : Read carefully, eliminate impossible options, and use common sense with logical rules.

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2. Statement and Conclusion

- Given a statement, identify which conclusion(s) logically follow.
- Use only the given information, no external knowledge.

Example :
Statement : All engineers are problem solvers. Some problem solvers are leaders.

Conclusions :
I. Some engineers are leaders. ✓ (Follows)
II. All leaders are engineers. ✗ (Does not follow)

3. Statement and Assumption

- Identify the assumption(s) that must be true for the statement to hold.
- Assumption is an unstated belief.

Example :
Statement : "Online education is becoming more popular among students."

Possible Assumption :
I. Students have access to the internet. ✓
II. Traditional education is ineffective. ✗

4. Statement and Argument

- Identify which argument is strong or weak in supporting/opposing the statement.
- A strong argument is logical, relevant and significant.

Example :
Statement : "Should school timing be changed to a later hour?"
Argument I : It will improve students' health and concentration. (Strong ✓)
Argument II : It will increase the burden on teachers. (Weak ✗)

5. Cause and Effect

- Identify the cause(s) and effect(s) in a given statement.
- Cause leads to effect. Check the direction of the relationship.

Example :
Statement : "Sales of electric vehicles have increased in the country."

Possible Cause :
I. Government incentives on EVs. ✓
II. Petrol prices have decreased. ✗

6. Analogies

- Based on a relationship between two words/numbers, find the missing term.
- Relationship can be of type: part-whole, function, category, cause-effect, etc.

Example :
Pen : Write :: Knife : ?
Options : (A) Cut (B) Eat (C) Metal (D) Sharp
Ans : (A) Cut (Function)

7. Series (Number, Letter, Mixed)

- Identify the pattern and find the next term.
- Pattern can be based on arithmetic, geometric, positional or mixed logic.

Example :
2, 6, 12, 20, 30, ?
Differences : +4, +6, +8, +10
Next difference = +12
Next term = 30 + 12 = 42
Ans : 42

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8. Coding - Decoding

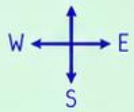
- Find the pattern in the code and apply the same logic.
- Can be based on position, reverse order, substitution, etc.

Example :
If CAT is coded as DBU, then DOG is coded as ?
Logic : Each letter is shifted by +1.
D → E O → P G → H
Ans : EPH

9. Direction Sense Test

- Based on movement and direction (North, South, East, West).
- Use diagrams to track position.

Example :
A person walks 10 m north, then 5 m east, then 10 m south. In which direction is he now from the starting point?
Net movement : 5 m East
Ans : East



10. Blood Relations

- Identify the relationship using family tree or logic.
- Use symbols or diagrams for clarity.

Example :
If A is the brother of B, and B is the daughter of C, how is A related to C?
Ans : A is the son of C.

11. Seating Arrangement

- People are arranged in a row, circle or other patterns with given conditions.
- Draw a diagram and place step by step.

Example :
Five people A, B, C, D, E are sitting in a row. B is to the right of A, C is to the left of D, E is at one end. Find the possible arrangement.
Approach : Use conditions one by one and eliminate invalid cases.

12. Tips & Common Mistakes

- ✓ Read the question carefully.
- ✓ Identify the type of reasoning question.
- ✓ Use diagrams, tables or symbols.
- ✓ Eliminate impossible options.
- ✓ Don't make assumptions beyond given data.
- ✓ Be systematic and organized.
- ✓ Practice previous year questions.
- ✓ Improve speed with regular practice.

Consistent
Practice
Real Progress !

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Small
Steps
Big Success !

Understand
Visualize
Solve
Score More !

9. Set Theory + Venn Diagrams + Counting

Sets
Diagrams
Counting
Stronger Aptitude !

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1. Basic Concepts of Sets

- A set is a well-defined collection of distinct objects.
- Elements are written inside { }.
- A = {a, b, c} means a, b, c are elements of set A.
- Notation : $x \in A$ (x is in A)
  $x \notin A$ (x is not in A)
- $|A|$ = number of elements in set A (cardinality)

Example :
$A = \{1, 2, 3, 4\}$ $\quad |A| = 4$
$5 \notin A$, $\quad 3 \in A$

2. Types of Sets

- Empty set ($\emptyset$) : No elements.
- Singleton set : One element.
- Finite set : Limited number of elements.
- Infinite set : Unlimited elements.
- Equal sets : A = B if they have same elements.
- Subset : $A \subseteq B$ if every element of A is in B.
- Proper subset : $A \subset B$ ($A \subseteq B$ and $A \neq B$)
- Universal set (U) : Set of all possible elements.

Example :
$A = \{1, 2\}$, $\quad B = \{1, 2, 3\}$
$A \subset B$ $\quad A \neq B$

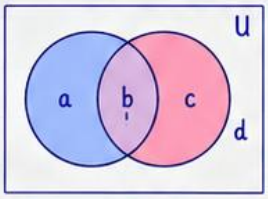
3. Set Operations

- Union : $A \cup B$ = elements in A or B (or both)
- Intersection : $A \cap B$ = elements common in A and B
- Difference : $A - B$ = elements in A but not in B
- Complement : A' = elements not in A (w.r.t. universal set U)

Example :
$A = \{1, 2, 3\}$, $\quad B = \{3, 4\}$
$A \cup B = \{1, 2, 3, 4\}$
$A \cap B = \{3\}$
$A - B = \{1, 2\}$
$B - A = \{4\}$

4. Venn Diagrams

- Used to visually represent sets and their relationships.
- Circles represent sets, overlapping region shows intersection.
- Useful for solving problems quickly.



a = only A
b = A ∩ B
c = only B
d = neither A nor B
U = a + b + c + d

5. Formulas (Two Sets)

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A - B) = n(A) - n(A \cap B)$
- $n(A') = n(U) - n(A)$
- If A and B are disjoint ($A \cap B = \emptyset$), then $n(A \cup B) = n(A) + n(B)$

Example :
$n(A) = 40$, $n(B) = 30$, $n(A \cap B) = 10$
$n(A \cup B) = 40 + 30 - 10 = 60$
$n(A - B) = 40 - 10 = 30$
$n(B - A) = 30 - 10 = 20$

6. Formulas (Three Sets)

- $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
- Useful for problems involving three groups.

Example :
$n(A) = 50$, $n(B) = 40$, $n(C) = 30$
$n(A \cap B) = 15$, $n(B \cap C) = 10$
$n(C \cap A) = 12$, $n(A \cap B \cap C) = 5$
$n(A \cup B \cup C) = 50 + 40 + 30 - 15 - 10 - 12 + 5 = 88$

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7. Counting Principles

- **Addition Principle** : If an event can occur in m ways or n ways (mutually exclusive), total ways = m + n.
- **Multiplication Principle** : If an event consists of two steps, with m ways for first and n ways for second, total ways = m × n.

Example :
A shirt can be chosen in 4 ways and a pant in 3 ways. Total outfits = 4 × 3 = 12

8. Inclusion-Exclusion Principle

- Used to count elements in union of multiple sets.
- For two sets :
  $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- For three sets :
  $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

When to use ?
When elements can belong to more than one set and overlapping is allowed.

9. Special Counting Techniques

- Counting distinct arrangements.
- Circular arrangements : $(n - 1)!$
- Counting with restrictions (at least, at most, exactly).
- Using complement method : Total − Unwanted cases.

Example :
Number of 4-digit numbers with no repetition using digits 0-9
$= 9 \times 9 \times 8 \times 7 = 4536$
(First digit cannot be 0)

10. Quick Tricks

- ✓ Draw Venn diagram for better visualization.
- ✓ Use complement when the condition is "not".
- ✓ Check for mutually exclusive cases.
- ✓ Use symmetry to reduce calculations.
- ✓ For "at least one", use inclusion-exclusion.
- ✓ In exam, try solving with numbers first.

11. Common Mistakes

- ✗ Forgetting to subtract intersection.
- ✗ Confusing $A - B$ with $A \cap B$.
- ✗ Wrong use of complement (always w.r.t U).
- ✗ Double counting elements.
- ✗ Not considering all cases in counting.
- ✗ Arithmetic mistakes in large numbers.

12. Exam Tips

- ✓ Identify sets and conditions carefully.
- ✓ Draw diagrams for complex questions.
- ✓ Use formulas directly to save time.
- ✓ Check if sets are disjoint or overlapping.
- ✓ Practice previous year questions.
- ✓ Keep track of units and total count.

Practice
More Questions
Build Confidence !

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Small
Steps
Big Results !

Observe
Practice
Apply
Get Better !

10. Geometry + Mensuration + Basic Statistics

Numbers
Shapes
Patterns
Real World !

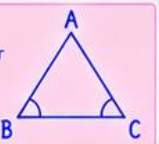
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1. Geometry - Basics

- Study of shapes, sizes, angles and properties of figures.
- Types : Point, Line, Line Segment, Ray, Angle, Triangle, Quadrilateral, Circle, etc.
- Important concepts : Parallel lines, Perpendicular lines, Angle properties, Triangles, Polygons, Circles.

Example :
In a triangle, the sum of all interior angles = 180°

$\angle A + \angle B + \angle C = 180°$



2. Angles & Triangles

- Types of angles : Acute (0° - 90°), Right (90°), Obtuse (90° - 180°), Straight (180°), Reflex (> 180°)
- Triangle properties :
  - Sum of angles = 180°
  - Exterior angle = sum of two interior opposite angles
  - Pythagoras theorem (Right triangle) : $a^2 + b^2 = c^2$

Example :
In right triangle, if base = 3 and height = 4, then hypotenuse = ?

$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$

3. Quadrilaterals & Polygons

- Types of quadrilaterals : Square, Rectangle, Parallelogram, Rhombus, Trapezium, Kite.
- Sum of interior angles of a polygon = $(n-2) \times 180°$
- Each exterior angle (regular polygon) = $\frac{360°}{n}$

Example :
Find the sum of interior angles of a hexagon (n = 6).

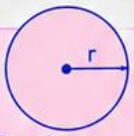
Sum = $(6 - 2) \times 180°$
= $4 \times 180° = 720°$



4. Circle - Basic Concepts

- Important terms : Radius (r), Diameter (d), Circumference, Chord, Arc, Sector, Segment, Tangent, Secant.
- Formulas :
  - Circumference = $2\pi r$ (or πd)
  - Area = πr^2
  - Angle at centre = 2 × (Angle at circumference on same arc)

Example :
If radius r = 7 cm, find circumference and area.



Circumference = $2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm

Area = $\pi r^2 = \frac{22}{7} \times 7^2 = 154$ cm²

5. Mensuration - 2D Figures (Area & Perimeter)

- Common Formulas :

Figure	Area	Perimeter
Square	a^2	$4a$
Rectangle	$l \times b$	$2(l + b)$
Triangle	$\frac{1}{2} \times b \times h$	$a + b + c$
Parallelogram	$b \times h$	$2(a + b)$
Trapezium	$\frac{1}{2}(a + b) \times h$	$a + b + c + d$
Circle	πr^2	$2\pi r$

Example :
Find the area of a rectangle with length = 8 m and breadth = 5 m.

Area = $8 \times 5 = 40$ m²

6. Mensuration - 3D Figures (Surface Area & Volume)

- Common Formulas :

Figure	Curved/Total Surface Area	Volume
Cube	$6a^2$	a^3
Cuboid	$2(lb + bh + hl)$	$l \times b \times h$
Cylinder	$2\pi r(r + h)$	$\pi r^2 h$
Cone	$\pi r(r + l)$	$\frac{1}{3}\pi r^2 h$
Sphere	$4\pi r^2$	$\frac{4}{3}\pi r^3$
Hemisphere	$3\pi r^2$	$\frac{2}{3}\pi r^3$

Example :
Find the volume of a cylinder with r = 3 cm, h = 10 cm.

Volume = $\pi r^2 h = \frac{22}{7} \times 3^2 \times 10 = \frac{22}{7} \times 9 \times 10 = \frac{660}{7}$ cm³

7. Coordinate Geometry (Basics)

- Cartesian plane with x and y axes.
- A point is represented as (x, y).
- Distance between two points :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Midpoint of a line segment :

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Example :
Find the distance between (1, 2) and (4, 6).

$d = \sqrt{(4 - 1)^2 + (6 - 2)^2} = \sqrt{3^2 + 4^2} = 5$

8. Basic Statistics - Key Terms

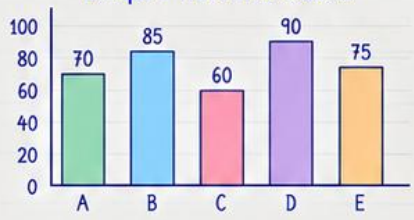
- Population : Complete set of observations.
- Sample : Subset of population.
- Data types :
  - Qualitative (Categorical)
  - Quantitative (Numerical)
- Measures : Mean, Median, Mode, Range, Variance, Standard Deviation.

Example :
Data : 4, 6, 8, 10, 12
Mean = $\frac{(4 + 6 + 8 + 10 + 12)}{5} = \frac{40}{5} = 8$
Median = 8 (middle value)
Mode = No mode (all occur once)
Range = 12 - 4 = 8

9. Data Representation

- Types : Tabular form, Bar graph, Histogram, Pie chart, Line graph.
- Used to summarize and visualize data.

Example : Marks of 5 Students



10. Important Formulas to Remember

- ✓ Area and perimeter formulas (2D figures).
- ✓ Surface area and volume formulas (3D figures).
- ✓ $\pi \approx 22/7$ or 3.14 (as per question).
- ✓ Sum of angles in triangle = 180°.
- ✓ Distance and midpoint formula.
- ✓ Mean, Median, Mode, Range.
- ✓ Practice previous year questions regularly.

11. Common Mistakes

- ✗ Forgetting units (cm, m, cm², m³).
- ✗ Using wrong formula for the figure.
- ✗ Confusing CSA and TSA.
- ✗ Calculation errors with π.
- ✗ Not checking if the question asks for area or perimeter / volume.
- ✗ Ignoring given conditions.
- ✗ Rounding off too early in calculations.

12. Tips & Tricks

- ✓ Draw neat diagrams.
- ✓ Write formulas before solving.
- ✓ Check units in the answer options.
- ✓ Use approximations only when allowed.
- ✓ Learn and practice standard values (π, √2, √3).
- ✓ Solve previous year questions.
- ✓ Improve speed with regular practice.

Practice
Build Concepts
Score High !

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Small
Steps
Big Results !

One Sheet Many Problems Big Confidence !

11. Quick Formula Sheet + Shortcuts + Tricks

Shortcuts Save Time Tricks Win Exams !

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1. Percentage, Ratio, Proportion, Averages

- Percentage : (Part / Whole) × 100
- Profit % : (Profit / Cost Price) × 100
- Loss % : (Loss / Cost Price) × 100
- Ratio $a : b = \frac{a}{b}$
- Proportion : $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$
- Average : $\frac{\text{Sum of values}}{\text{Number of values}}$

Trick : For x% of y → y% of x = same value.

2. Profit & Loss + Simple & Compound Interest

- Profit % : $\frac{(SP - CP)}{CP} \times 100$
- Loss % : $(CP - SP) \times 100$
- Simple Interest : $SI = \frac{PRT}{100}$
- Amount (SI) : $A = P + SI$
- Compound Interest : $A = P\left(1 + \frac{R}{100}\right)^n$
  $CI = A - P$

Trick : For 2 years CI, use $\left(1 + \frac{r}{100}\right)^2$ directly (faster).

3. Time, Speed & Distance + Work & Pipes

- Speed = $\frac{\text{Distance}}{\text{Time}}$
- Distance = Speed × Time
- Time = $\frac{\text{Distance}}{\text{Speed}}$
- Work : Total Work = Efficiency × Time
- Pipes : Net rate = Inlet − Outlet

Trick : Use units (km/h, m/s) to avoid mistakes.

4. Algebra + Equations + Progressions

- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a - b)^2 = a^2 - 2ab + b^2$
- $a^2 - b^2 = (a - b)(a + b)$
- Linear Eq. : $ax + b = 0 \Rightarrow x = -\frac{b}{a}$
- Quadratic : $ax^2 + bx + c = 0$
  $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- AP : $a_n = a + (n - 1)d$
- GP : $a_n = ar^{n-1}$
- Sum of AP : $S_n = \frac{n}{2}[2a + (n - 1)d]$
- Sum of GP : $S_n = \frac{a(r^n - 1)}{r - 1}$ $(r \neq 1)$

Trick : In options, try plugging numbers to save time.

5. Permutation, Combination + Probability

- ${}^nP_r = \frac{n!}{(n - r)!}$
- ${}^nC_r = \frac{n!}{r!(n - r)!}$
- Probability : $P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$
- $0 \leq P(E) \leq 1$
- $P(E) + P(\overline{E}) = 1$

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Trick : For "at least one" → use 1 − P(none).

6. Data Interpretation + Data Sufficiency

- Percentage = $\frac{\text{Part}}{\text{Whole}} \times 100$
- Growth % = $\frac{(\text{New} - \text{Old})}{\text{Old}} \times 100$
- Average = $\frac{\text{Sum}}{\text{Count}}$
- Ratio = $a : b = \frac{a}{b}$
- Data Sufficiency :
  - S1 alone sufficient
  - S2 alone sufficient
  - Both together sufficient
  - Each alone not sufficient

Trick : Read the question first, then data/graph.

7. Logical Reasoning + Analytical Reasoning

- Use Venn diagrams, tables, and patterns.
- Types : Series, Coding-Decoding, Blood Relations, Direction Sense, Seating Arrangement, Cause-Effect, Syllogism.
- Look for keywords : always, never, some, all, only.
- Eliminate impossible options.

Trick : Draw a small diagram, it reduces confusion.

8. Set Theory + Venn Diagrams + Counting

- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- $n(A - B) = n(A) - n(A \cap B)$
- $n(A') = n(U) - n(A)$
- For three sets :
  $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$
- Counting : Use multiplication principle and casework.

Trick : In questions, draw a Venn diagram to avoid mistakes.

9. Geometry + Mensuration + Basic Statistics

- Triangle : Area = $\frac{1}{2}$ × base × height
- Circle : Area = πr^2 , Circumference = $2\pi r$
- Rectangle : Area = $l \times b$, Perimeter = $2(l + b)$
- Cube : TSA = $6a^2$, Volume = a^3
- Cuboid : TSA = $2(lb + bh + hl)$, V = $l \times b \times h$
- Cylinder : CSA = $2\pi rh$, TSA = $2\pi r(h + r)$
- Sphere : SA = $4\pi r^2$, V = $\frac{4}{3}\pi r^3$
- Statistics : Mean = $\frac{\sum x}{n}$,
  Median (odd) = $\frac{n+1}{2}$ th term

Trick : Learn standard values − $\pi \approx \frac{22}{7}$ or 3.14 , $\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$

10. Important Approximations & Values

- $\sqrt{2} \approx 1.41$ $\sqrt{3} \approx 1.73$ $\sqrt{5} \approx 2.24$
- $\pi \approx \frac{22}{7}$ or 3.14
- Common squares : $1^2 = 1$ $2^2 = 4$ $3^2 = 9$ $4^2 = 16$ $5^2 = 25$ $10^2 = 100$
- Common cubes : $1^3 = 1$ $2^3 = 8$ $3^3 = 27$ $5^3 = 125$ $10^3 = 1000$
- % to fraction : 50% = $\frac{1}{2}$, 25% = 1/4 12.5% = 1/8 , 20% = 1/5

Trick : Memorize small squares and cubes (1-20).

11. Shortcuts & Tricks

- ✓ Use options to eliminate (ODR).
- ✓ Plug in simple numbers (1, 2, 0, −1).
- ✓ Use unit digit method in large calculations.
- ✓ For % change, use $\frac{(\text{New} - \text{Old})}{\text{Old}} \times 100$.
- ✓ In averages, use assumed mean method.
- ✓ In mixture problems, use alligation.
- ✓ For time & work, use LCM.
- ✓ Draw diagrams for reasoning questions.
- ✓ Use approximation when exact value not required.
- ✓ Keep a list of common values and formulas.

12. Golden Exam Tips

- ✓ Learn and revise formulas regularly.
- ✓ Practice previous year questions.
- ✓ Use shortcuts to save time.
- ✓ Do not over-calculate.
- ✓ Check units, signs and conditions.
- ✓ Manage time effectively.
- ✓ Attempt easy questions first.
- ✓ Keep a separate formula sheet for revision.
- ✓ Stay calm and confident.
- ✓ Practice more, improve accuracy.

Formulas Shortcuts Higher Scores !

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Study Smart Solve Fast You Can Do It !

Practice Previous Year Solve More !

12. GATE Aptitude Previous-Year Patterns + Top Practice Questions

Same Questions Better You !

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1. GATE Pattern (General Aptitude)

- General Aptitude (GA) = 15 marks
- Total Questions = 10
- 5 questions × 1 mark
- 5 questions × 2 marks
- Topics : Verbal / English, Quantitative Aptitude, Analytical / Logical Reasoning, Data Interpretation.

Note : Exact topic-wise distribution may vary every year. Focus on all areas.

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2. Common Question Styles

- Verbal Ability (vocabulary, grammar, sentence completion, analogy, word usage)
- Quantitative Aptitude (numbers, ratios, percentages, time-speed-work, etc.)
- Analytical / Logical Reasoning (logical statements, arrangements, coding-decoding, cause-effect, etc.)
- Data Interpretation (tables, bar graphs, pie charts, line graphs, etc.)

3. Previous-Year Patterns (Recurring)

- Ratios, Percentages, Averages
- Time, Speed, Distance, Work
- Counting, Permutations & Combinations, Probability
- Data Interpretation (tables / graphs)
- Set Theory, Venn Diagrams
- Geometry, Mensuration (basic)
- Logical Statements, Cause-Effect
- Arrangements (seating, scheduling)
- Analogy, Vocabulary, Odd One Out
- Grammar, Sentence Correction/ Completion

Note : These are common patterns, not a fixed distribution. Questions may vary.

4. Important Formulas (Quick Revise)

- Average : $\bar{x} = \frac{\text{Sum}}{n}$
- Percentage : $\frac{\text{Part}}{\text{Whole}} \times 100$
- Time, Speed, Distance : Speed = $\frac{D}{T}$
- Work : Total Work = Efficiency × Time
- Probability : $P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$
- Sets : $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Tip : Don't just memorize, understand when to apply.

5. Top Practice Questions

Q1 The average of 5 numbers is 24. If one number is removed, the average of the remaining 4 numbers is 20. Find the removed number.

(A) 20 (B) 24
(C) 32 (D) 40

Ans : 40

Q2 A : B = 3 : 5 and B : C = 10 : 7. Find the ratio A : C.

(A) 6 : 7 (B) 7 : 6
(C) 3 : 7 (D) 5 : 7

Ans : 6 : 7

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Q3 A train covers 360 km in 4.5 hours. What is the average speed of the train?

(A) 60 km/h (B) 70 km/h
(C) 80 km/h (D) 90 km/h

Ans : 80 km/h

Q4 Two fair dice are rolled. What is the probability that the sum of the numbers is 9 ?

(A) 1/6 (B) 1/8
(C) 1/9 (D) 1/12

Ans : 1/9

Q5 In a class of 60 students, 35 know C, 30 know Python and 15 know both. How many students know at least one of the two languages?

(A) 35 (B) 45
(C) 50 (D) 55

Ans : 50

Q6 All engineers are graduates. Some graduates are researchers. Which of the following conclusion(s) necessarily follows ?

(A) All engineers are researchers
(B) Some researchers are engineers
(C) All engineers are graduates
(D) Some engineers are researchers

Ans : (C) All engineers are graduates

6. Quick Strategy

1. Read the question carefully.
2. Identify the topic.
3. Estimate/approximate the answer before calculating.
4. Eliminate incorrect options.
5. Watch out for units, signs and logical traps.
6. Don't round off too early.
7. Practice previous year questions regularly.

Tip : Speed + Accuracy = Higher Score !

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7. Exam Checklist

- ☑ Attempt easy questions first.
- ☑ Manage time (don't spend too long on one question).
- ☑ Keep a separate rough sheet.
- ☑ Recheck calculations if time allows.
- ☑ Solve previous year papers and mock tests.

Consistency Leads to Results !

GATE Prep Today Better Tomorrow !

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Small Steps Big Results !